# 🎯 QR Attendance Feature - Test Guide

---

## Part 1: Understanding Event Types

### Mandatory Event

- Attendance is **required**
- Has **positive hours** (rewarded for attending)
- Has **negative hours** (penalty for being absent)
- Penalty can be applied by admin to deduct hours from absent students

### Non-Mandatory Event

- Attendance is **optional**
- Has positive hours only
- No penalty for missing the event

---

## Part 2: Adding & Deleting Rolls

### Non-Mandatory Event

| Action | Expected Result |
|---|---|
| Add a student | Student's total hours should **increase** by the event's positive hours |
| Add same student again | Should be **rejected** - student already marked present |
| Delete a student | Student's total hours should **decrease** by the event's positive hours |

### Mandatory Event (Before Penalty is Applied)

| Action | Expected Result |
|---|---|

| | |
|---|---|
| Add a student | Student gains positive hours<br>Student is marked present → will NOT get penalty later |
| Delete a student | Student loses positive hours<br>Student is marked absent → WILL get penalty when applied |

### Mandatory Event (After Penalty is Applied) ⚠️

This is the critical scenario. Pay close attention!

| Action | Expected Result |
|---|---|
| **Add a student** | 1. Student gains positive hours<br>2. If student was penalized earlier, they should **regain** the negative hours<br>**Net effect: +positive hours + regained penalty** |
| **Delete a student** | 1. Student loses positive hours<br>2. Student should now **receive** the penalty (lose negative hours)<br>**Net effect: -positive hours - penalty applied** |

---

## Part 3: Editing Events

### A. Changing Mandatory Status

#### Mandatory → Non-Mandatory

| Penalty Status | Expected Result |
|---|---|
| Before penalty applied | No immediate impact on students |
| After penalty applied | ⚠️ **All penalized students should regain their negative hours** |

#### Non-Mandatory → Mandatory

| Scenario | Expected Result |
|---|---|
| Event has attendees | Attendees keep their hours; now eligible to apply penalty to absentees |
| Apply penalty | All absent students (in relevant wings) should lose negative hours |

### B. Changing Wings

| Change | Expected Result |
|---|---|
| **Add a wing** | Students in new wing become "relevant" to the event<br>If penalty was applied → newly included absent students should be penalized |
| **Remove a wing** | Students in removed wing become "irrelevant"<br>If they were penalized → they should regain their penalty<br>If they attended → they keep their positive hours |

### C. Changing Positive Hours

| Change | Expected Result |
|---|---|
| Increase hours (e.g., 2 → 4) | All attendees should gain the difference (+2 hours) |
| Decrease hours (e.g., 4 → 2) | All attendees should lose the difference (-2 hours) |

**Note:** This should affect ALL students who have already attended.

### D. Changing Negative Hours

| Change | Before Penalty Applied | After Penalty Applied |
|---|---|---|
| Increase (e.g., -1 → -2) | No immediate impact | Penalized students lose additional hours |
| Decrease (e.g., -2 → -1) | No immediate impact | Penalized students regain the difference |
| Set to 0 | No penalty possible | All penalized students regain full penalty |

---

## Part 4: Wing-Based Behavior

### Key Rules

1. Penalty should **ONLY** apply to students who are in the event's target wing(s)
2. Students not in the event's wing should NOT receive penalty even if they're absent
3. If a student attends an event from a different wing, they still get the positive hours

### Scenarios

| Situation | Expected Result |
|---|---|
| Student in matching wing, attended | Gets positive hours |
| Student in matching wing, absent | Gets penalized (if mandatory + penalty applied) |
| Student in different wing, attended | Gets positive hours (attendance counts) |
| Student in different wing, absent | NO penalty (they're not in target wing) |

---

## Part 5: Test Scenarios

### Scenario 1: Basic Attendance

1. Create a non-mandatory event with 2 hours
2. Add Student A
3. Delete Student A
4. **Verify:** Student A's hours return to original value

### Scenario 2: Mandatory Event Flow

1. Create mandatory event with 3 hours, -1 negative hours
2. Add Student A (present)
3. Leave Student B absent
4. Apply penalty
5. **Verify:** Student A has +3 hours, Student B has -1 hour

### Scenario 3: Adding Roll After Penalty

1. Follow Scenario 2
2. Now add Student B to the event
3. **Verify:** Student B should now have +3 hours (gained positive + regained penalty)

### Scenario 4: Deleting Roll After Penalty

1. Follow Scenario 2
2. Now delete Student A from the event
3. **Verify:** Student A should now have -1 hour (lost positive + received penalty)

### Scenario 5: Changing to Non-Mandatory

1. Follow Scenario 2
2. Edit event: change to non-mandatory
3. **Verify:** Student B should regain the -1 penalty

### Scenario 6: Wing Change Impact

1. Create mandatory event for "Wing A" with 2 hours, -1 negative
2. Add Student A (from Wing A) - present
3. Student B (from Wing A) is absent
4. Apply penalty
5. Edit event: add "Wing B"
6. Student C (from Wing B) was not present
7. **Verify:** Should Student C now be penalized?

### Scenario 7: Hours Change

1. Create event with 3 hours
2. Add Student A (gets 3 hours)
3. Edit event: change to 5 hours
4. **Verify:** Does Student A now have 5 hours or still 3?

---

## Part 6: Questions to Consider

1. **Double Marking Prevention:** How do you prevent the same student from being marked twice?
2. **Event Status Check:** Should attendance be allowed on a closed (non-live) event?
3. **Delete Confirmation:** Should there be a confirmation before deleting attendance?
4. **Hours Recalculation:** When event details change, how do you efficiently recalculate hours for all affected students?
5. **Wing Validation:** How do you determine which wing a student belongs to?
6. **Audit Trail:** Should you track who added/deleted attendance and when?
7. **Edge Cases:**
   - What if negative hours is greater than positive hours?
   - What if a student has no wing assigned?
   - What if the event has no wings selected?

## Summary: Quick Reference

| Event Type | Add Roll | Delete Roll | Edit Hours | Edit Negative | Change Type | Change Wing |
|---|---|---|---|---|---|---|
| Non-Mandatory | +hours | -hours | Update all | N/A | See rules | N/A for penalty |
| Mandatory (before penalty) | +hours | -hours | Update all | No impact | See rules | See rules |
| Mandatory (after penalty) | +hours, +regain | -hours, -penalty | Update all | Adjust penalties | See rules | Adjust penalties |

Good luck with your implementation! 🚀